Syed Ahsan Masud Zaidi

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SUMMARY

- Ph.D. candidate in Computer Science researching **computer vision, vision foundation models, and video understanding**, with research experience in **offline reinforcement learning and robotic control**.
- Developed **training-free contact localization, video action recognition, and reproducible GPU workflows** using **Python and PyTorch**, integrating grounded detection, video segmentation, and motion analysis.
- Authored research papers in **IEEE/CVF CVPR Workshops (CVSports), ICPR, and IEEE/RSJ IROS**, with applications in **sports medicine and athlete safety**.

PROFESSIONAL EXPERIENCE

Kansas State University 2023–Present

Doctoral Researcher | *Computer Vision and Video Understanding*

- **Grounded perception (GRAZE):** Built a training-free pipeline combining **Grounding DINO**, **SAM2 mask propagation**, and motion-aware temporal refinement to localize player-dummy contact in untrimmed video.
- **Video action recognition:** Adapted pretrained **ViViT** for tackling-technique classification using contact-centered clips, expert annotations, and **imbalance-aware training**.
- **Reproducible ML:** Built training and evaluation workflows for **ViViT, MViTv2, SlowFast, and VideoMAE v2**, with **session-disjoint splits and leakage checks** to assess generalization.
- **ML data (TackleNet):** Released expert-annotated practice videos with privacy preprocessing, fixed splits, and reusable benchmark software.
- **Robot learning:** Coauthored **CRISP (IROS 2026)** on **context-robust skill inference for offline reinforcement learning**.

Kansas State University 2023–Present

Graduate Teaching Assistant | *Instructor of Record, CIS 209, since 2026*

- Teach **Python and C** in **CIS 209** through lectures, coding labs, and individual debugging support.
- Develop course materials and assessments; **mentor teaching assistants** and give feedback on programming assignments.

University of Engineering and Technology Lahore 2014–2023

Lecturer | *Mechatronics and Control Engineering*

- Taught undergraduate **programming, embedded systems, digital logic, signals and systems, and electronics**.
- Directed the **Simulation Lab** and served as **In-Charge of the Semester Cell**.

RBL PepsiCo 2013–2014

Project Engineer

- Commissioned a **KRONES PET filling plant**, supporting system integration and production handover.

TECHNICAL SKILLS AND SOFTWARE

Programming and ML: Python, C, C++, MATLAB; PyTorch, TensorFlow, scikit-learn, Hugging Face Transformers.

Computer vision: Action recognition, event localization, object detection, segmentation, OpenCV, RetinaFace.

Foundation models: Grounding DINO, SAM2, SAM 3.1, pretrained video models, zero-shot inference, model fine-tuning.

Evaluation and computing: Class-imbalance handling, cross-validation, session-disjoint evaluation; Git, Linux, SLURM, GPU training on Beocat and PSC Bridges-2.

Robotics and engineering: ROS, control systems, signal processing, embedded systems, LabVIEW.

EDUCATION

Kansas State University 2023–Present
Ph.D. Candidate in Computer Science | GPA: **3.96/4.0**. Advisor: Lior Shamir.
Research: computer vision, video understanding, and AI for sports safety.

King Fahd University of Petroleum and Minerals 2019
M.S. in Electrical Engineering | Dhahran, Saudi Arabia.
Thesis research: laser-beam position control for free-space optical communication.

University of Engineering and Technology Lahore 2013
B.S. in Mechatronics and Control Engineering | Lahore, Pakistan.
Thesis: Static Marker-Based Cricket Bowling Analysis.

SELECTED PUBLICATIONS

1. **GRAZE: Grounded Refinement and Motion-Aware Zero-Shot Event Localization**
   Syed Ahsan Masud Zaidi, Lior Shamir, William Hsu, Scott Dietrich, and Talha Zaidi.
   IEEE/CVF CVPR Workshops, Computer Vision in Sports (CVSports), 2026. Spotlight. *Paper*

2. **ViTs for Action Classification in Videos: An Approach to Risky Tackle Detection in American Football Practice Videos**
   Syed Ahsan Masud Zaidi, William Hsu, and Scott Dietrich.
   International Conference on Pattern Recognition (ICPR), 2026. *Paper* | *Publisher*

3. **CRISP: Context-Robust Skill Inference for Long-Horizon Offline Robotic Control**
   Talha Zaidi, Arslan Munir, and **Syed Ahsan Masud Zaidi**.
   IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2026. *Code*

4. **Reproducing Vision Transformer-Based Risky Tackle Detection in American Football Practice Videos: A Companion Study**
   Syed Ahsan Masud Zaidi, Lior Shamir, and Scott Dietrich.
   RRPR Workshop at ICPR, 2026. Oral presentation; Springer LNCS post-proceedings forthcoming. *Conference listing*

OPEN SOURCE RESEARCH SOFTWARE

TackleNet benchmark resources Dataset | Code
Released a PyTorch dataset loader, fixed split files, baselines, and shared evaluation utilities for technique classification and contact localization. Used RetinaFace and SAM 3.1 for face de-identification.

GRAZE inference workflow Code
Provided inference scripts, environment setup, SLURM batch jobs, and prediction aggregation to support reproducible contact-localization experiments.

TECHNICAL LEADERSHIP AND PROFESSIONAL SERVICE

- **Research communication:** Presented a CVPR Workshops spotlight and poster, an ICPR main-conference poster, and an RRPR oral presentation in 2026.
- **Peer review:** IEEE Access; SLAS Technology; Reproducible Research in Pattern Recognition (RRPR) at ICPR; Women in Computer Vision (WiCV) at ECCV.

AWARDS AND RECOGNITION

- **AOSSM Playmaker Grant Lab team finalist**, American Orthopaedic Society for Sports Medicine, 2026.
- **CVPR Broadening Participation support**, Computer Vision Foundation, 2026.
- **IAPR Travel Stipend and Registration Scholarship**, ICPR 2026.
- **Full Merit Scholarship**, M.S. studies at KFUPM.